User Interaction modeling with Generative AI

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Abstract

This research introduces a modified T5 transformer model designed to make digital conversations feel more natural by focusing on emotional context. The work includes: (1) building a data pipeline for the EmpatheticDialogues dataset that labels text using Plutchik's eight primary emotions; (2) updating the T5 architecture with a specialized emotion embedding layer to help the model process feelings alongside words; and (3) training the system to understand and generate empathetic responses at the same time. By blending emotional data directly into the model's internal logic, responses are more sensitive than standard methods. This technology shows great promise for use in therapy bots and social robots.

1 Introduction

As AI becomes a bigger part of daily lives, the need exists for systems that do not just process facts, but also understand human feelings. This is especially vital for tools used in mental health, elderly care, and social robotics. While current large language models are great at writing clear sentences, the emotional vibe of a conversation is often missed. This can result in responses that feel robotic, cold, or even tone deaf during sensitive moments.

Most traditional systems focus on grammar and logic, but human talk is driven by emotion. If a robot gives a factually correct but emotionally flat answer to someone who is upset, trust is broken. Current methods usually just stick an emotion label like [happy] onto the text (Zhou et al., 2018; Deng and Ren, 2023), but this is a surface level fix. It does not change how the model actually thinks about each word generated (Radford et al., 2018).

In high stakes areas like caregiving, models must do more than just spot an emotion. Adaptation of the entire speaking style to match the user's state is required (Hempel et al., 2023). To solve this, emotional data is integrated directly into the model's core structure using Plutchik's framework (Plutchik, 1980). By adding emotion at the token level, the model adjusts tone more naturally throughout a sentence, leading to much more human like interactions.

2 Prior Research

Early attempts at creating empathetic AI focused on basic emotion detection (Spezialetti et al., 2020; Szabóová et al., 2020). Key studies used models like BERT for understanding context and GPT for generating text, but neither could perfectly balance emotional depth with fluent speech (Rashkin et al., 2019). Older systems also relied on static word embeddings like Word2Vec or GloVe, which are emotion blind. These systems might treat "I am fine" and "I am struggling" as similar because the sentence structure is alike (Mikolov et al., 2013; Pennington et al., 2014; Devlin et al., 2019).

Some researchers tried to fix this by adding an extra training step to make word embeddings more sensitive to feelings (Seyeditabari et al., 2019). More recent complex systems use multiple lexicons and memory banks to help models like LLaMA 2 handle therapy sessions (Rasool et al., 2025). However, these can be slow and hard to scale for real time use.

The approach used here is different because it is simpler and more flexible. Emotional vectors are baked directly into the T5 model (Raffel et al., 2019). By using the Transformer architecture (Vaswani et al., 2017) and BART style logic (Lewis et al., 2020), the system shifts emotional tone dynamically. If no emotion is detected, a neutral setting is used, making the system robust for real world use without needing constant manual input.

3 Technical Approach

The model's emotional intelligence is improved using a clear four step strategy. First, the EmpatheticDialogues dataset is prepared by cleaning the text and adding specific emotion tags. Next, the T5 model's code is modified to include a custom emotion layer. Plutchik's Wheel (see Figure 1) is used to categorize every input, ensuring the model sees the feeling behind the words.

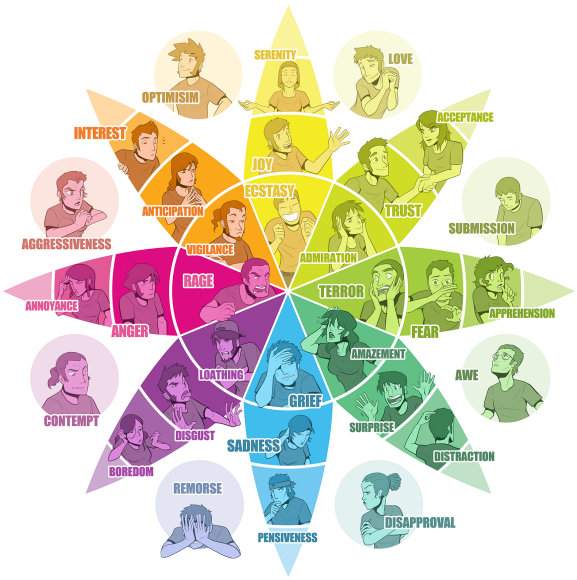

Figure 1: Plutchik's emotional wheel

During the training phase, this emotion augmented T5 is fine tuned so it can learn the relationship between sentiment and sentence structure. When the model generates a response, both the meaning of the words and the emotional cues provided are considered. Finally, the work is checked using standard AI metrics and custom emotional tests to ensure the responses are both accurate and caring. Figure 2 shows the overall workflow.

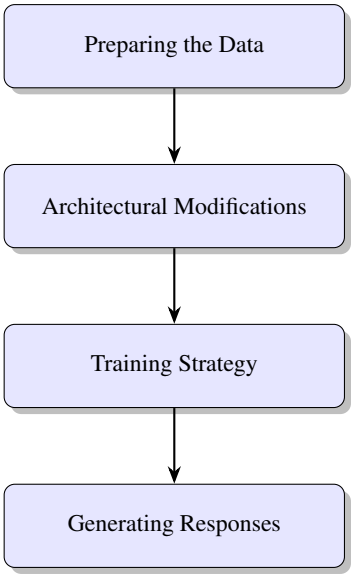

Figure 2: Overview of the proposed methodology

3.1 Preparing the Data

The EmpatheticDialogues dataset is used because of its focus on real emotional exchanges. Text is cleaned and multi turn conversations are turned into simple input response pairs. To label the emotions, a pre trained DistilBERT classifier is used (Sanh et al., 2019; Demszky et al., 2020) and results are mapped to Plutchik's eight categories: joy, trust, fear, surprise, sadness, disgust, anger, and anticipation. This provides a rich dataset where every conversation has a corresponding emotional tag.

3.2 Architectural Modifications

To give the T5 model a sense of feeling, a new class called `T5WithEmotionEmbeddings` is created. This adds a layer where each emotion is represented as a math vector that matches the model's internal size. When the model processes a sentence, the emotion vector is added to the word vector. This ensures the model's thoughts are shaped by the emotional context from the very start. If an emotion is not specified, a neutral setting is used to keep things stable.

3.3 Training Strategy

The model is trained using a custom `EmotionAwareTrainer` based on the Hugging Face library (Wolf et al., 2020). Unlike standard training that only looks at word accuracy, this trainer checks if the generated text matches the required emotional tone. This forces the model to learn how different feelings change the way

people speak, resulting in much more empathetic outputs.

3.4 Generating Responses

The final system uses a function called `generate_response()`. When a user speaks, the system identifies the emotion, picks a matching prompt template, and then uses the T5 model to craft a reply. By combining the specific emotion ID with the text, the model generates a response that is not just a generic answer, but one that actually fits the user's mood.

4 System Performance

Several methods are used to see how well the system works. During training, the loss scores which track errors are monitored to make sure the model is actually learning and not just memorizing. Both the training and testing scores drop steadily, which is a sign of a healthy functioning model.

To measure the quality of the chat, standard AI tools like BLEU (Papineni et al., 2002), ROUGE L (Lin, 2004), and BERTScore (Zhang et al., 2020) are used. These look at how similar the AI's answers are to real human answers. A SentenceTransformer (Reimers and Gurevych, 2019; Wang et al., 2020) is also used to filter the results, ensuring that the AI stays on topic while maintaining the right emotional tone.

5 Voice Interaction Features

A speech to speech loop is also added so users can actually talk to the AI. This creates a hands free natural experience that feels more like a real conversation.

5.1 How the Voice System Works

- **Listening:** `Wav2Vec2` (Baevski et al., 2020) is used to turn spoken audio into text.
- **Feeling:** The system analyzes that text to find the emotion using a GoEmotions trained classifier.
- **Thinking:** The T5 model generates a caring response based on those feelings.
- **Speaking:** The `pyttsx3` library is used to read the response out loud in a clear voice.

5.2 The User Experience

A simple visual interface is built using `Tkinter`. Users click "Start Recording," speak for up to 50 seconds, and then see the transcription and the AI's detected emotion on the screen while hearing the reply.

6 Experimental Results

The following tables show how the T5 model performs against human references.

6.1 BLEU Scores

BLEU scores measure how many words match between the AI and a human. As shown in Table 1, the scores are modest, which is common for creative dialogue models that do not just copy paste answers.

BLEU Score	Value
BLEU 1	0.1509956890511106
BLEU 2	0.04790281091809629
BLEU 3	0.021323349433213956
BLEU 4	0.012320208411349908
Average BLEU	0.058135514453442685

Table 1: BLEU Scores for Emotion Aware T5 Model

6.2 ROUGE Scores

ROUGE scores focus on how well the model summarizes or maintains sequence. Table 2 shows that the model is decent at keeping the main points but still has room to improve flow.

ROUGE Score	Value
ROUGE 1	0.1411779721654755
ROUGE 2	0.01653054298642534
ROUGE L	0.12909146130255003
ROUGE Lsum	0.1294476906549279

Table 2: ROUGE Scores for Emotion Aware T5 Model

6.3 Semantic Meaning

The BERTScore is 0.86, which is quite high. This means that even if the AI did not use the exact same words as a human, the meaning of the response is very close.

7 Development Challenges

Building this system involved several difficulties. Issues with computer power, model choice, and data quality were encountered.

7.1 Hardware and Computing

GPT 2 and DistilGPT were originally tried, but they were too slow. Nearly 20 hours were required to train just one version, and any small error meant starting over. Because unlimited access to high end GPUs was not available, the shift to T5 was made, which is much more efficient.

7.2 Choosing the Right Model

GPT models read text in one direction, making it hard to force an emotion into the middle of a sentence. T5's encoder decoder structure is a better fit. It allows emotional context to be fed into the encoder before the decoder starts speaking, leading to much better results.

7.3 Data Gaps

The EmpatheticDialogues dataset is a great start, but it is mostly short chats. It does not cover complex or mixed emotions. To make a truly smart AI, datasets that include longer more complicated human interactions are needed.

8 Current Limitations

There are still a few hurdles to clear before this AI is perfect.

8.1 Learning Rare Emotions

The model is better at common feelings like joy because more data exists for them. Rare feelings like disgust or fear are harder for the AI to learn (see Figure 3).

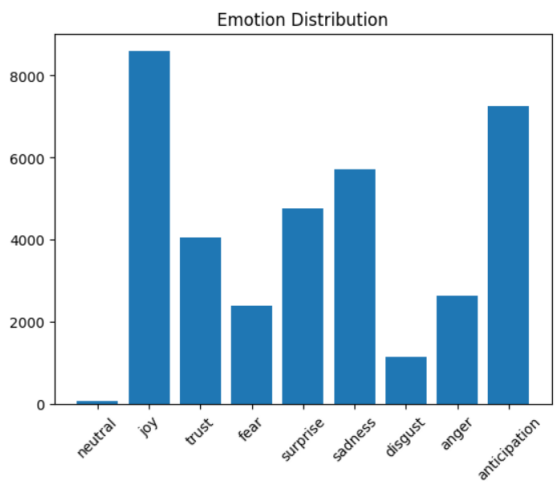

Figure 3: Emotion Distribution in the Dataset

8.2 Real World Generalization

Since only one main dataset is used, the AI might struggle with topics not seen before. It also does not handle blended emotions well yet, as only one primary feeling is picked for every sentence.

8.3 Better Testing Needed

Metrics like BLEU and ROUGE are just math. These metrics do not actually know if a response felt empathetic. In the future, humans are needed to grade the responses to see if the AI is truly connecting with people.

9 Future Roadmap

Plans exist to make this system better, including using more datasets like DailyDialog (Li et al., 2017) and MELD (Poria et al., 2019). A move toward soft labeling is desired, where the AI can feel multiple emotions at the same time (Liapis et al., 2025).

Other goals include:

- **Multilingual Support:** Using models like mT5 (Xue et al., 2020) to help the AI speak other languages.
- **Personalization:** Allowing the AI to remember user details and past conversations.
- **Speed:** Making the model smaller and faster so it can run on a phone or a small robot.

10 Summary

A T5 based model is successfully created that uses emotion embeddings to have more caring conversations. By teaching the AI to process feelings as part of core logic, the project moves beyond simple text matching toward true digital empathy. While work remains on data balance and complex feelings, this project proves that AI can be built to understand not just what is said, but how it is felt. These findings are a step forward for social robots and digital mental health support.

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